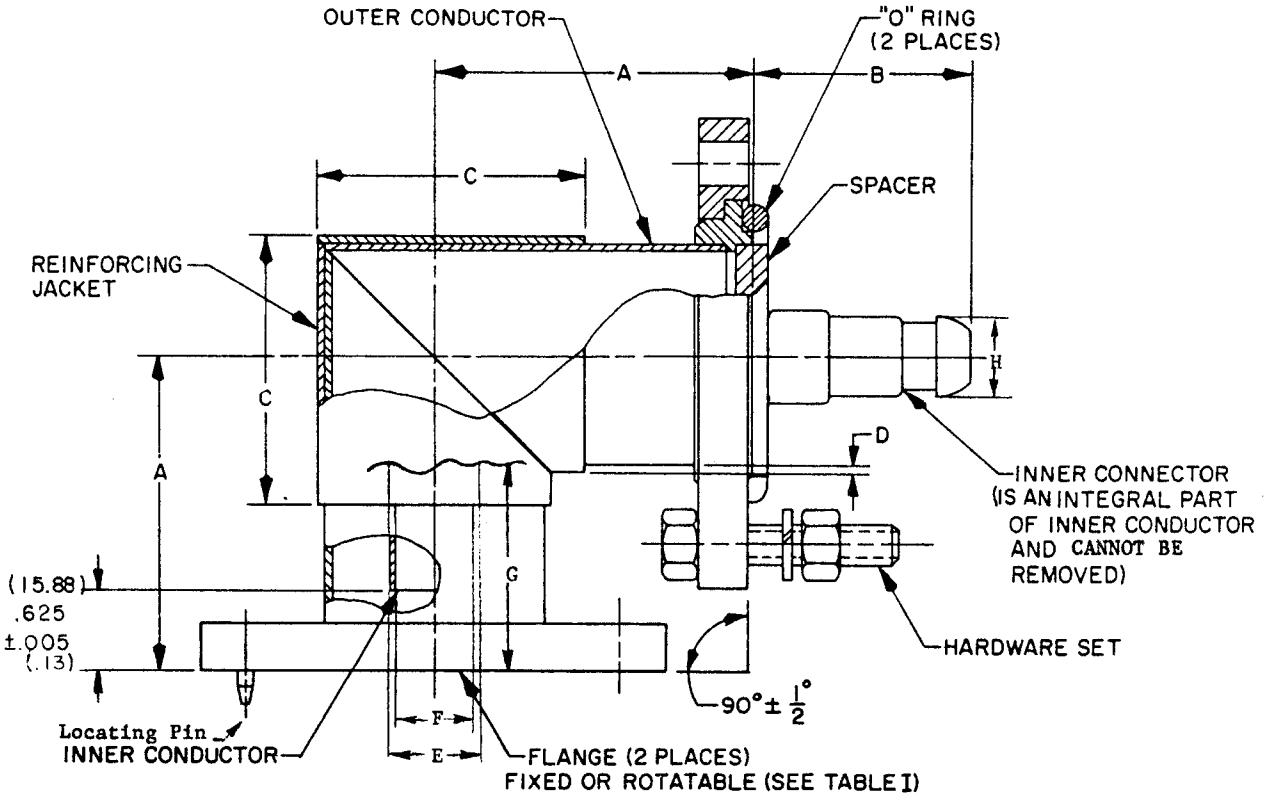
MILITARY SPECIFICATION SHEET

LINE ASSEMBLIES, RADIO FREQUENCY TRANSMISSION 1-5/8 INCH ELBOWS

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the elbow assemblies described herein shall consist of this specification and the latest issue of MIL-L-28796.



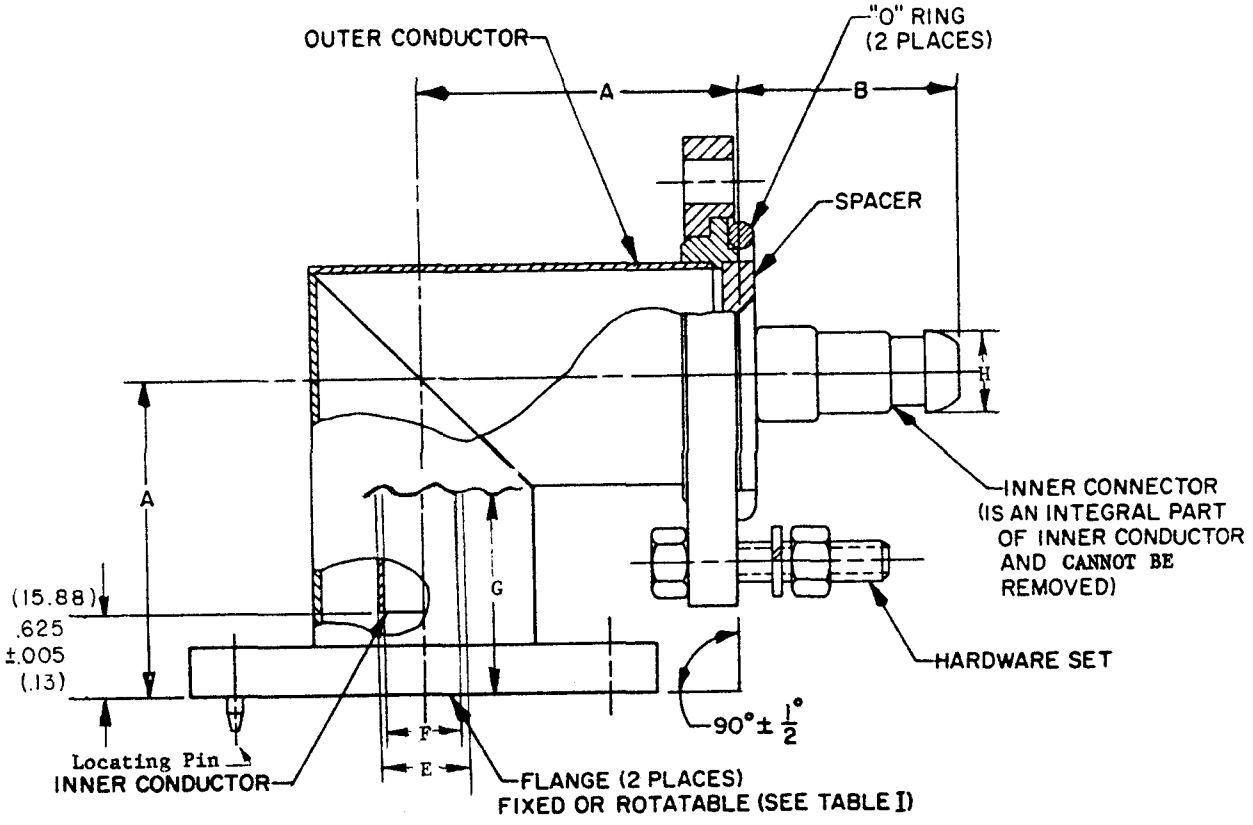
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. 90 degree miter copper elbow assemblies.

Ⓐ denotes changes

FSC 5985

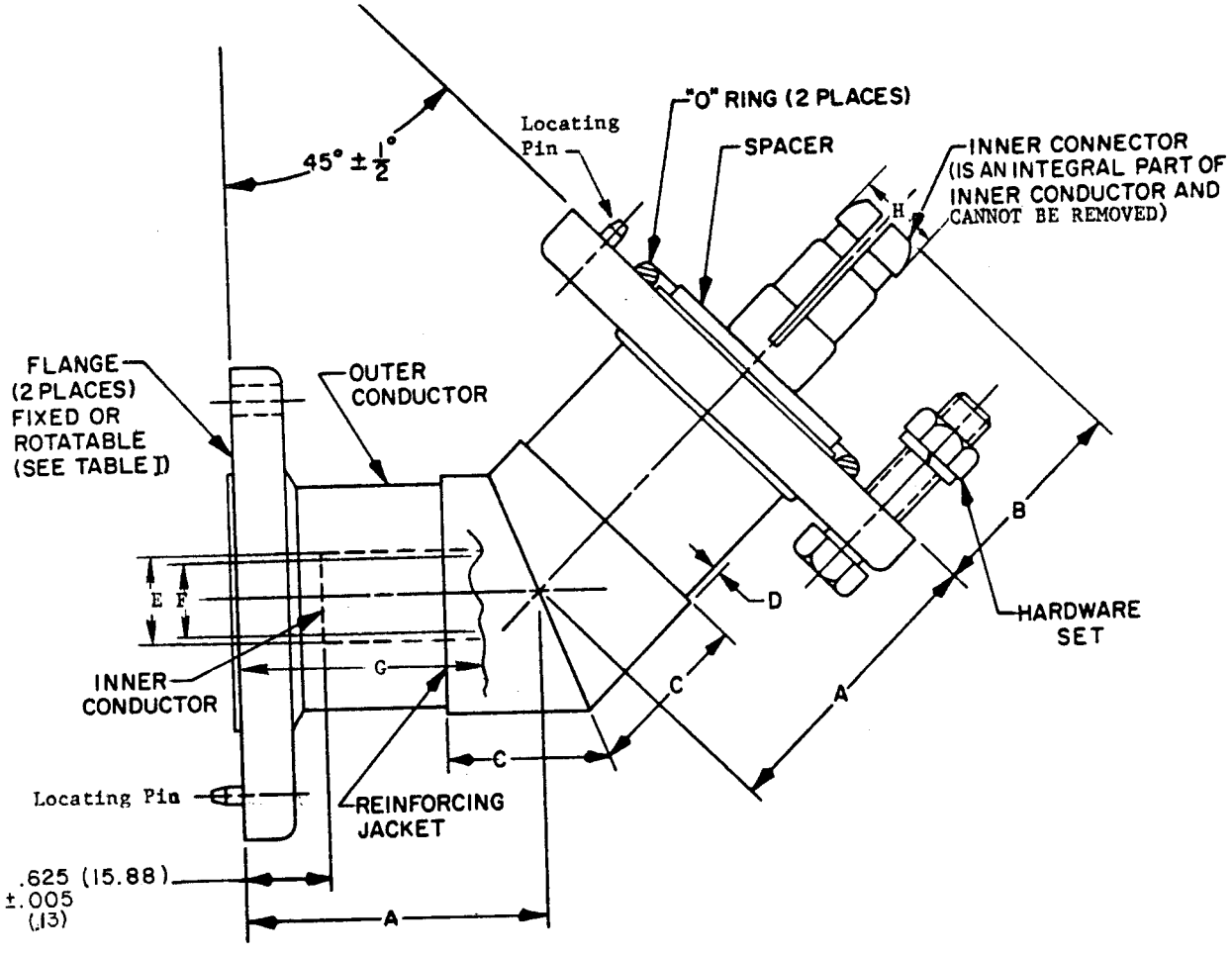
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. 90 degree miter aluminum elbow assemblies.

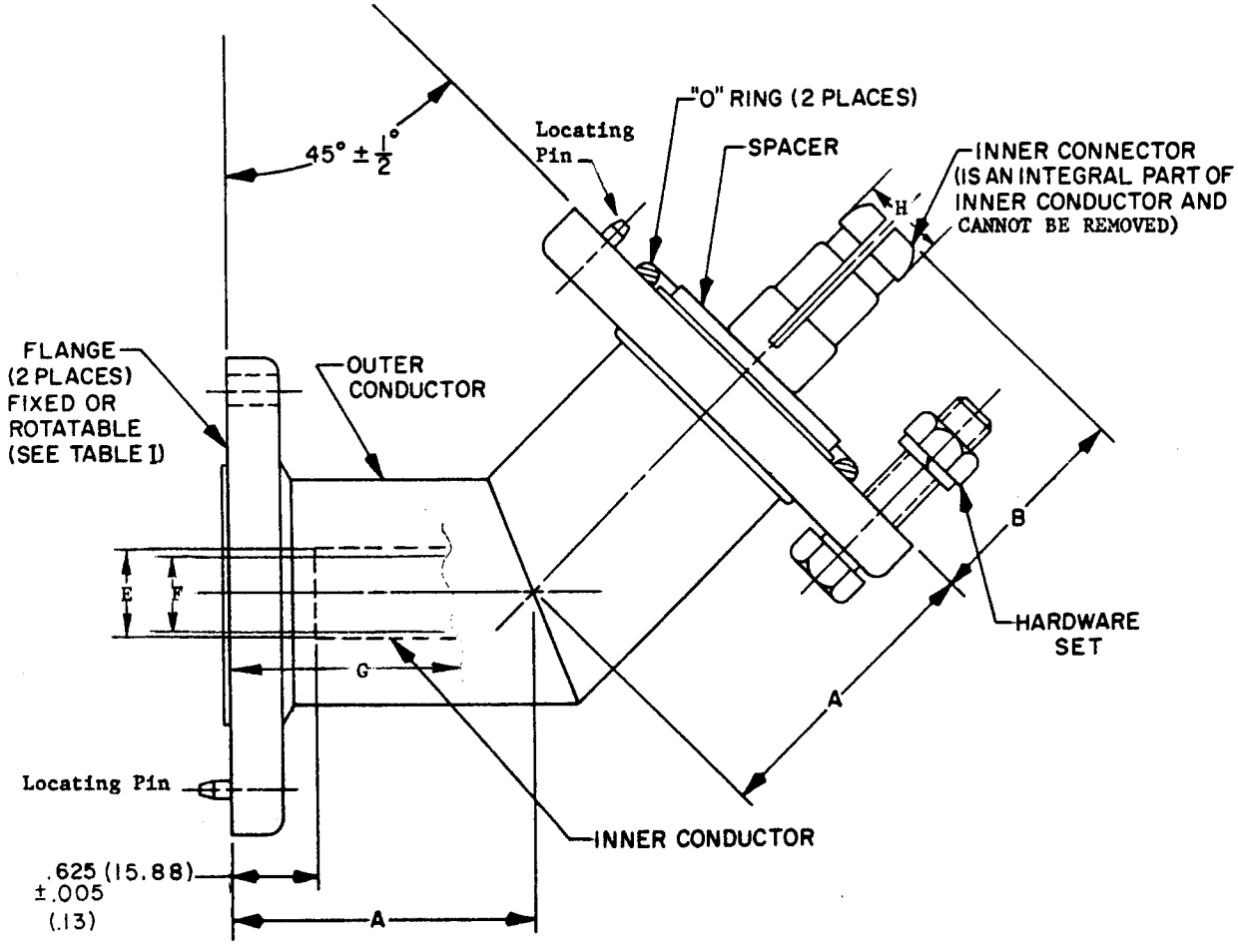
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 3. 45 degree miter copper elbow assemblies.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
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FIGURE 4. 45 degree miter aluminum elbow assemblies.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 through 4 and table 1.

Operating frequency: 400 to 450 MHz.

(A) Weight (approximate): 3 1/2 lbs.

Gaskets: Class 2b, grade 50 of ZZ-R-765.

(A) Hardware set: Four each, corrosion-resistant steel bolts, nuts, and lockwashers, .3125-18UNC, 1-1/2 inches long.

Finish:

Inner conductor - Silver plated per QQ-S-365, type I.

Outer conductor and flanges (aluminum) - Chemical film per MIL-C-5541, type II, grade C, class 3.

Surface roughness: 63 microinches.

Insertion loss: .1 dB maximum.

VSWR: 1.02:1 maximum.

Impedance: 50 ±.3 ohms.

Velocity of propagation: 99.7 percent minimum.

RF Power handling capability:

Average - 10.0 kilowatts minimum.

Peak - 136 kilowatts minimum.

Dielectric withstanding voltage: 10 kilovolts (ac).

Pressurization:

Test - Gage pressure of 65 (+2, -0) lbf/in^2.

Operating - Gage pressure of 20 ±2 lbf/in^2.

Part number: M28796/1-(and dash number from table I).

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TABLE I. Dash numbers and characteristics.

Dash number	Flange per MIL-F-24044/	Transmission line per MIL-L-3890 and type	Inner connector per MIL-F-24044/	Dimensions								Figure number
				A ±.016	B ±.016	C ±.031	D +.010 -.000	E ±.0025	F ±.0025	G 3/ min	H +.003 -.005	
001	1-05 (fixed)	1-RG-153/U	1-05C	2.375	1.625	1.975	.050	.6640	.5880	1.641	.590	1
002	2-05 (rotatable)	1-RG-153/U	2-05C	2.375	1.625	---	---	.6640	.5880	1.641	.590	1
003	1-06 (fixed)	3-RG-382/U 2/	1-05C	2.375	1.625	1.975	.050	.6640	.5880	1.641	.590	2
004	2-06 (rotatable) 1/	3-RG-382/U 2/	2-05C	2.375	1.625	---	---	.6640	.5880	1.641	.590	2
005	1-05 (fixed)	1-RG-153/U	1-05C	2.375	1.625	1.975	.050	.6640	.5880	1.641	.590	3
006	2-05 (rotatable)	1-RG-153/U	2-05C	2.375	1.625	---	---	.6640	.5880	1.641	.590	3
007	1-06 (fixed)	3-RG-382/U 2/	1-05C	2.375	1.625	1.975	.050	.6640	.5880	1.641	.590	4
008	2-06 (rotatable) 1/	3-RG-382/U 2/	2-05C	2.375	1.625	---	---	.6640	.5880	1.641	.590	4

1/ The same as MIL-F-24044/2-06, except dimension D for figure 5A of MIL-F-24044/2 shall be Min - 1.773, Nom - 1.773 and Max - 1.783.

2/ The same as RG-382/U of MIL-L-3890/3 except the outer conductor outside diameter shall be 1.77 ±.01 and the outer conductor wall thickness shall be 0.1215.

3/ Distance which straight section of inner conductor with specified inside and outside diameter shall extend to allow insertion of inner connector.

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